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Machine Teaching for Explainable AI in Industry: A Novel Approach for Time Series Classifiers

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Abstract

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Acknowledgements

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Chapter 1

Introduction

1.1 Background and Motivation

Artificial intelligence (AI) has become a central tool in modern decision-making across industry, healthcare, and society at large. Systems based on deep learning, probabilistic models, and other machine learning approaches are now embedded in critical applications such as disease diagnosis, fraud detection, industrial process control, and energy management. What these domains have in common is the demand for high accuracy and efficiency, but also for transparency. If the reasoning of an AI system cannot be understood, it becomes difficult to trust, audit, or improve. This concern has been one of the main driving forces behind the increased interest for *Explainable AI (XAI)*, a field dedicated to providing human-understandable explanations for AI outputs [1, 3].

While XAI research has achieved progress in areas such as image and text classification, *time series data* presents unique difficulties. Unlike images, which can be visually inspected, or text, which humans interpret naturally through language, time series are abstract sequences of values over time. The meaning of their patterns is often domain-specific and not intuitive to non-experts [35, 46].

These examples below have been selected from a wide array of domains in order to illustrate this challenge.

- **Healthcare**: Electrocardiograms (ECGs) record a heart's electrical activity over time. Cardiologists learn to recognise anomalies that indicate e.g. arrhythmias, but for a layperson, the waveforms are little more than lines on a graph [19].

- **Finance**: Stock price developments are notoriously hard to predict. The high dimensionality of the data, coupled with complex causal relationships, make it difficult for even experienced analysts to isolate which temporal changes matter most in predicting future trends [41].
- **Cybersecurity**: Intrusion detection relies on monitoring network traffic patterns. Anomalous activity may only be detectable by subtle deviations in timing or frequency, which humans struggle to recognise unaided [5].
- **Energy systems**: Load curves representing electricity consumption evolve across days and weeks, where unusual fluctuations may signal fraud, inefficiency, or faults [46].
- **Transportation**: Vehicle telemetry, including engine vibrations, brake pressure, and GPS trajectories, generates multivariate time series where abnormal events may be hidden within noisy signals [28].

In all these settings, humans are poorly equipped to parse long, multi-dimensional sequences. People tend to focus on a few salient peaks or drops, and may overlook less obvious but crucial dependencies. This cognitive mismatch means that black-box time series classifiers (TSCs) risk becoming tools whose outputs are accepted or rejected without true understanding [47].

There are several reasons why time series are particularly difficult for human interpretation:

1. **Temporal length**: Sequences often contain hundreds or thousands of points. Unlike images, where the whole object is visible at once, time series must be "read" across time, which strains working memory [29].

2. **Multivariate complexity**: Many applications involve multiple signals recorded simultaneously, e.g., power, state-of-charge, and ambient temperature in electric vehicle (EV) charging. The interplay between these variables can be non-linear and unintuitive.

3. **Lack of semantic anchors**: In text, words carry meaning, and in images, shapes can be labelled. In time series, the same rise or dip may have different implications depending on context. For example, as batteries approach full charge, they become harder to charge and the power output from the charger tends to taper off. However, should the same pattern occur at 40% charge, it might indicate a fault.

4. **Hidden dependencies**: Important relationships may occur across distant time steps. Humans struggle to track these long-range dependencies without computational assistance [22].

The cognitive limitations above lend credit to the idea of generating simplified example-based explanations for TSC interpretability [35, 21].

One direction for interpretability is to highlight which features, time steps, or subsequences drive a classifier's decision. For images, saliency maps show which pixels influence an output. For time series, however, generating equivalent explanations is far less straightforward. Some methods adapt feature attribution techniques such as SHAP or LIME, while others identify discriminative subsequences called *shapelets* or employ counterfactual perturbations [20]. These methods tend to tell a fragmented picture, as users are shown isolated points or segments without a coherent story of why the classifier reached its decision.

1.1.1 MT4XAI: Example-based explanations through machine teaching

The research project *Machine Teaching for Explainable AI (MT4XAI)* aims to address these interpretability challenges by reframing explanations of AI behaviour as a teaching problem [45, 52, 14]. Instead of overwhelming users with all possible examples or post-hoc rules, MT4XAI identifies a small, optimised *teaching set* of examples that illustrate the concept a model has learned.

This builds on the theory of *machine teaching*, which studies how a teacher can teach a concept to a learner as efficiently as possible using the most informative examples [48]. In the MT4XAI framework, the AI system plays the role of the "teacher", while the "learner" is the human user, and the "concept" to be taught is the decision boundary of a TSC. The challenge is to find teaching sets that are both simple and faithful: small enough for humans to grasp, but accurate enough to reflect the model's reasoning.

This master's thesis aims to contribute to MT4XAI by applying the framework to a real-world case study in collaboration with *Eviny*, a major operator of EV charging stations in Scandinavia. The project develops an anomaly detection system for EV charging sessions and explanation mechanisms that leverage recent advances in time series simplification and machine teaching. By simplifying complex charging curves and selecting

optimised teaching sets, the thesis aims to provide explanations that are both accurate and cognitively accessible. This practical case study serves as a validation of MT4XAI's theoretical ideas and as a step toward their adoption in industrial practice.

1.2 Problem Statement

Deep learning models have proven effective for time series classification/anomaly detection tasks, but their opacity limits adoption in high-stakes applications such as energy, health care and transport. Traditional XAI methods are not well suited to time series, as they often fail to produce explanations that are faithful to the model while also being accessible to a wide audience of non-experts. Anomaly detection systems for EV charging cannot provide customers or domain experts with actionable insights without justifying their classifications with explanations. This creates a gap between predictive performance and practical usability.

1.3 Research Questions and Objectives

The overarching research question for this project is formulated as follows:

> **"How can techniques from machine teaching for XAI be applied to time series classifiers in order to generate simple, understandable and faithful explanations of model decisions in a real world industry setting?"**

The user study will be designed to answer the following hypotheses:

- H1: Participants who receive ORS-based simplifications will classify abnormal sessions more accurately than those who see raw time series.
- H2: Teaching sets ordered by a simplicity prior (curriculum learning) will improve human learners' ability to simulate the decisions of a Time Series Classifier on EV charging data, compared to unordered teaching sets.

TEKSTALTERNATIV 1: To answer the research question and test the hypotheses, the project pursues the following objectives:

1. **Wrangle Data**: Load, clean, and enrich the EV charging dataset.
2. **Analyse**: Conduct exploratory data analysis (EDA) to analyse the statistical characteristics and limitations of the dataset.
3. **Model**: Design, train, tune, and evaluate multiple machine learning models for time series forecasting.
4. **Detect Anomalies**: Select a model for the anomaly detection pipeline. Implement, evaluate, and select the overall anomaly detection approach by integrating different error metrics.
5. **Simplify**: Adapt and evaluate simplification techniques such as ORS.
6. **Test**: Design and test a machine teaching system that provides example-based explanations to users.

TEKSTALTERNATIV 2: In order to answer the research question and test the hypothesis, we will need to complete a series of objectives. The first objective is to load, clean, and enrich the EV charging dataset. The second objective will be to learn about the statistical characteristics and limitations of the dataset by conducting an exploratory data analysis (EDA). The modelling phase involves selecting an appropriate model architecture and reliable error metrics for modelling, and to train, tune (find optimal hyperparameters) and evaluate multiple deep learning models. The fourth objective is to implement, evaluate and select the overall anomaly detection algorithm and error metric. We then adapt and evaluate simplification techniques such as ORS, before finally designing and testing a machine teaching system that provides example-based explanations to users.

1.4 Thesis Structure

The remainder of this thesis is organised as follows:

- Chapter 2 introduces the theoretical background, covering time series classification, explainable AI, machine teaching, and learnability theory.
- Chapter 3 reviews related work, focusing on XAI methods for time series, simplification algorithms, and recent contributions in machine teaching for XAI.

- Chapter 4 describes the methodology, including data collection, preprocessing, model design, simplification algorithms, evaluation metrics, and the proposed MT4XAI pipeline.
- Chapter 5 presents model and simplification experiments, reporting on anomaly detection performance, simplification quality, and teaching set selection.
- Chapter 6 details the implementation of the MT4XAI tool, including system architecture and web application design.
- Chapter 7 reports on the human evaluation study, describing its objectives, design, procedure, and results.
- Chapter 8 provides a discussion of the findings, limitations, and contributions to the MT4XAI project and industry.
- Chapter 9 concludes the thesis and outlines directions for future work.

Chapter 2

Background and Theoretical Foundations

2.1 Time Series Data and Classification

Time series data are sequences of observations ordered in time. They occur in diverse domains such as healthcare, finance, energy, and transportation, and are characterised by temporal dependencies, autocorrelation, and often non-stationarity [29]. In the context of this thesis, the primary focus is on electric vehicle (EV) charging sessions, which generate multivariate time series capturing variables such as power output and the battery's state of charge (SOC) over time for each session. The EV charging dataset supplied by the Norwegian renewable energies company *Eviny* allows us to explore the practical challenges that arise from implementing an explainable anomaly detection system in a real-world industry context.

2.1.1 Techniques for Time Series Classification

A wide array of machine learning techniques has been applied to time series classification (TSC), ranging from traditional statistical models to modern deep learning architectures [22].

- Traditional models:

  - Logistic Regression: Estimates class probabilities, most effective for linearly separable time series.
  - Decision Trees: Rule-based and non-linear, able to capture complex interactions, but prone to overfitting.
  - k-Nearest Neighbours (kNN): Classifies a series by comparing it to its nearest neighbours; intuitive but less scalable in high dimensions.

- Deep learning models: These have achieved state-of-the-art results in TSC by learning hierarchical representations directly from raw sequences [23].
  - Convolutional Neural Networks (CNNs): Effective for extracting local patterns via convolutional filters.
  - Long Short-Term Memory (LSTM) networks: Recurrent models designed to capture long-range dependencies in sequential data.
  - Temporal Convolutional Networks (TCNs): Causal, dilated convolutions that capture long-range dependencies without recurrence.
  - Transformers: Attention-based models that flexibly capture global dependencies across a sequence.

  These models provide strong predictive performance but vary in interpretability. In particular, deep learning models are often criticised as "black boxes," motivating the use of explainability techniques.

Many real-world applications involve multivariate forecasting. This involves predicting future values of several interdependent variables simultaneously. For example, forecasting both charging power (kW) and SOC trajectories requires modelling their complex interactions. Static features such as ambient temperature or charger nominal power can also be incorporated, as they causally affect battery chemistry and charging behaviour.

The prediction horizon H defines how far into the future the model must forecast. Short-term horizons are generally easier to predict than long-term ones, as error compounds over time.

2.1.2 Anomaly Detection on Time Series Data

A common approach in time series anomaly detection is to train forecasting models on sequences considered normal. During deployment, the model's forecasts are compared to actual observations, and deviations beyond a threshold can indicate anomalies [8]. Deep

learning methods for time series anomaly detection span forecasting-, reconstruction-, and density-based approaches, among others [8, 50]

All approaches to anomaly detection must deal with the tension between statistical and perceptual anomalies. What a model flags as unusual based on high variance or error may not align with what humans perceive as surprising or suspicious. Ultimately, anomaly labels often carry a degree of subjectivity.

2.1.3 Error Metrics

The choice of error metric directly impacts what kinds of time series are classified as anomalous. We therefore include an overview of some commonly used metrics for time series forecasting, along with the specialised metrics we explore in this project.

- Mean Absolute Error (MAE):

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|y_i - \hat{y}_i|$$

- Mean Squared Error (MSE):

$$\text{MSE} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2$$

- Root Mean Squared Error (RMSE):

$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$$

- Huber Loss (SmoothL1): Quadratic for small errors, linear for large errors. This offers robustness against outliers.

$$L_\delta(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2, & \text{if } |y - \hat{y}| \leq \delta \\ \delta\big(|y - \hat{y}| - \frac{1}{2}\delta\big), & \text{otherwise} \end{cases}$$

- Weighted Macro Root Mean Squared Error (Macro-RMSE): Used for multivariate, multi-horizon settings. For each horizon h and channel $c \in \{\text{power}, \text{soc}\}$, the per-horizon RMSE is

$$\text{RMSE}_{h,c} = \sqrt{\frac{1}{T_{h,c}} \sum_{t=1}^{T_{h,c}} (y_{t,h,c} - \hat{y}_{t,h,c})^2}.$$

With horizon weights w_h (satisfying $\sum_{h=1}^{H} w_h = 1$, e.g. $w_h \propto e^{-\lambda(h-1)}$), the session-level Macro-RMSE is

$$\text{MacroRMSE} = \sum_{h=1}^{H} w_h \Big[w \cdot \text{RMSE}_{h,\text{power}} + (1-w) \cdot \text{RMSE}_{h,\text{soc}} \Big],$$

where w is the power weight and $\lambda \geq 0$ is the horizon-decay hyperparameter.

- Robust Weighted Standardised Error (RWSE): Residuals are first computed per time step, horizon, and channel:

$$r_{t,h,c} = y_{t,h,c} - \hat{y}_{t,h,c}.$$

From validation residuals, we compute the median $m_{h,c}$ and median absolute deviation $\text{MAD}_{h,c}$, then standardise:

$$z_{t,h,c} = \frac{|r_{t,h,c} - m_{h,c}|}{\text{MAD}_{h,c}},$$

with clipping (e.g. $z_{t,h,c} \leq 5.0$). To combine across horizons and channels, we apply horizon weights w_h (exponential decay, normalised to 1) and channel weights w_c (uniform, sum to 1). The session-level RWSE is

$$\text{RWSE} = \frac{1}{T'} \sum_{t=1}^{T'} \sum_{h=1}^{H} \sum_{c=1}^{C} w_h w_c \, z_{t,h,c},$$

where T' is the number of evaluated time steps, H is the number of horizons, and C is the number of target variables (or "channels"). When forecasting for *power (kW)* and *SOC*, $C = 2$.

Each metric emphasises different aspects of error: RMSE heavily penalises large spikes, MAE balances across magnitudes, Huber Loss introduces robustness to outliers, and RWSE highlights systematic deviations rather than isolated spikes.

2.2 Explainable Artificial Intelligence (XAI)

Explainable Artificial Intelligence (XAI) seeks to make the behaviour of learning systems intelligible to humans. In practice, this spans approaches that aim to produce interpretable models by design (e.g., sparsity- or prototype-based models), and post-hoc approaches that explain already-trained systems via local or global surrogates, feature attributions, examples, or counterfactuals [31, 13, 18]. Importantly, to generate explanations is more than just a technical challenge. Fundamentally, it is a human–AI interaction problem in which people form, evaluate and revise mental models of an AI system [36].

In the early 1970's, expert systems such as *MYCIN* included built-in explanation facilities that produced rule-based justifications and traces of inference in response to users' "why" and "how" questions [42, 9, 44]. After the deep learning resurgence in the 2010's, general-purpose post-hoc methods emerged, notably LIME [39] and SHAP [32], together with attribution techniques such as Integrated Gradients [43]. Around the same period, researchers called for conceptual clarity and scientific standards for interpretability [31, 13]. Some researchers highlighted the need for human-centred explanations by drawing on philosophy and the social sciences [36]. Good explanations are not just long lists of causes. They are contrastive ('why P rather than Q?'), selective rather than exhaustive, and inherently social. Such explanations should aim to address the user's specific question, context, and goals [36].

More recently, theory-driven accounts of how explanations shape human understanding have appeared. For example, Yang et al. propose a psychological theory of explainability that models user understanding as Bayesian updating over a similarity space, and validate it against human predictions of model output given saliency maps [49]. On the other hand, the visual analytics community highlights that trust is dynamic and partly non-technical, and that well-designed interactive visualisation can help build trust in AI systems [7].

2.2.1 Why XAI?

Demand for XAI arises from regulation and risk management, but also from practical needs to calibrate trust, debug models, and support accountability. Empirical work shows that explanations should be designed for the user task (e.g., auditing vs. learning a boundary) and that interactivity matters: letting users probe, compare alternatives, and

run “what-ifs” can support more accurate mental models, while poor visual or interaction design can undermine trust [7]. A psychological view makes this concrete: explanations guide belief-updating about the model. Effective explanations align with human inductive biases and contrastive questions [36, 49].

2.2.2 Families of XAI methods

Feature attribution techniques use gradient- and perturbation-based scores (saliency, integrated gradients, occlusion) to estimate how input features contribute to a prediction [51, 43]. They are widely used, but are sensitive to implementation choices and may be hard to interpret without extensive knowledge of the application domain / subject matter. [2, 16, 36, 47]

Other techniques such as LIME [39] have seen widespread usage and popularity due to their simplicity and interpetability. Local surrogates or global rule/tree-like surrogates can approximate a black-box model’s behaviour for a ”neighbourhood” of the input space with a simpler model [39, 18]. This technique can also be used for the whole input space, however, these explanations quickly lose fidelity [37].

Example-based explanations are a family of techniques that rely on prototypes (i.e. representative instances), criticisms, and counterfactuals that communicate decisions through concrete examples that are often more natural for people to reason with [37]. In time series, instance-based approaches—prototypes and counterfactuals—are increasingly used for model inspection and user studies [47, 15, 12, 4].

Concept-based methods explain models using human-understandable ideas (“concepts”) rather than raw features. TCAV learns a direction in a hidden layer that corresponds to a user-specified concept and measures its influence on the prediction, while related work either discovers concepts automatically (ACE) or builds them in as an explicit intermediate layer via concept bottleneck models. This enables inspection and edits [25, 17, 26].

Mechanistic interpetability “opens the box” to identify neurons or small subnetworks (“circuits”) that implement particular computations and trace how information flows through layers [6, 38, 11]. These circuit analyses are less common for time-series models than for vision and language. [47].

A more recent line of work explores the use of large language models (LLMs) to generate or mediate explanations. Because LLMs are trained on vast corpora of natural language, they can produce fluent, human-readable rationales for predictions, either for their own outputs or for other models' decisions [33, 10]. Early results suggest that LLMs can act as "explanation engines" by converting technical information into natural language or by serving as surrogates that generate counterfactuals and rationales [27, 24]. However, concerns remain about factuality and faithfulness, since LLMs may produce plausible, but misleading explanations - or hallucinate entirely.

2.2.3 Limitations and critiques

In recent years, valid concerns about the limitations of XAI methods have been raised. First, many technical metrics (e.g., "faithfulness") say little about whether users actually understand the model or make better decisions, as rigorous user-centred evaluation remains scarce and challenging/expensive [13]. Second, saliency and perturbation methods can be unstable, dataset-dependent, and prone to confirmation bias. Third, explanations can "over-persuade", creating unwarranted confidence. Finally, the social sciences caution that explanations are contrastive and selective, and that evaluation should reflect this rather than assume a single "complete" account that can capture the complexity of a high-capacity model [36].

2.2.4 XAI for time series models

Time series bring additional hurdles to the task of generating explanations. As previously discussed, long sequences strain working memory, multivariate signals interact in complex ways, and identical shapes can have different meanings under different temporal or physical contexts. Methods originally developed for images or tabular data usually aren't directly applicable [47].

The current literature clusters XAI techniques for time series classifiers (TSC) into three families [47]:

- *time-point-based* attributions (saliency over samples or channels)
- *subsequence-based* methods (e.g., shapelets, wavelet components)
- *instance-based* approaches using prototypes and counterfactuals.

Surveys and position papers also call for more rigorous evaluation protocols tailored to temporal data [40]. Systems providing post-hoc, model-agnostic explanations via prototypes and counterfactuals can be evaluated by detertmining to what extent users can forward-simulate a classifier. A recent study suggests that users can benefit for XAI for TSCs, but time series remain cognitively demanding and the human subjects' motivation to spend time on studying explainations strongly affect outcomes [20].

In an attempt to reduce the cognitive and visual complexity of time series explanations, recent work has explored simplifying the series before applying XAI methods [46]. Complementary work has proposed metrics for evaluating simplification algorithms more broadly, showing that simplified series can improve interpretability compared to raw inputs, particularly when datasets are seasonal, non-stationary, or low-entropy [34]. Together, these studies suggest that simplification can bridge the gap between model decisions and human perception without substantially sacrificing classification behaviour.

For explaining pure forecasting/regression models, much work adapts attributions and perturbations to temporal settings (e.g., meaningful perturbations over windows, profile-based reasoning). However, rigorous benchmarks and user studies lag behind classification, and instability under temporal perturbations is a recognised concern [40, 47].

2.2.5 Implications for this thesis

Three points from the literature shape our approach. First, explanations for time series classifiers should be contrastive and example-based, as these align with human reasoning and reduce cognitive load [36, 20]. Secondly, interactivity supports agency but must be carefully designed to avoid over- or under-trust [7]. Third, curve simplification tailored to the classifier can offer a principled path to cognitively accessible yet faithful explanations for TSC's [46].

2.3 Machine Teaching (MT) Fundamentals

2.4 Learnability Theory and Teaching Dimension

2.5 Summary and Research Gap

Chapter 3

Related Work

3.1 XAI for Time Series Classification

3.2 Time Series Simplification Algorithms

The Optimal Robust Simplifications (ORS) algorithm reduces a time series to a small set of straight-line segments while preserving the classifier's decision under perturbations, balancing fidelity, simplicity, and robustness [46].

3.3 Machine Teaching in XAI

3.4 Positioning of the Thesis in the MT4XAI Research Project

Chapter 4

Methodology

4.1 Data Collection and Preprocessing

4.1.1 EV Charging Session Data

4.1.2 Data Wrangling and Feature Engineering

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4.2 Time Series Classification Models for Anomaly Detection

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4.3.3 Hyperparameter Tuning

4.4 Machine Teaching for XAI Framework

4.4.1 Adapting MT4XAI Formalism

4.4.2 Teaching Set Determination

4.5 Evaluation Metrics

4.5.1 Anomaly Detection Macro-RMSE, RWSE, Horizon Decay

In this project, we have applied macro-averaging across horizons to ensure that no single prediction horizon dominates evaluation, which helps balance the models' short- and long-term predictive ability. This is particularly important in anomaly detection pipelines [30]. Specifically, we applied Huber Loss in the model training and validation loop, target-weighted macro-RMSE for the model selection stage, and evaluated both weighted macro-RMSE and RWSE for the inference/forecasting stage in the anomaly detection pipeline.

4.5.2 Simplification

4.5.3 User Study

4.6 End-to-End Pipeline & API

Chapter 5

Discussion

5.1 Interpretation of Key Findings

5.2 Limitations

5.3 Contribution to MT4XAI and Industry

Chapter 6

Conclusion and Future Work

6.1 Conclusion

6.2 Future Work

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Appendix A

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